

The Resolution Operator

Where H_5 comes from, why the five is not a choice, and the resolution at which it meets the floor

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WHAT THIS DOCUMENT SETTLES

The retention operator H_5 was written down from intuition about how numbers should divide over distance, with no derivation behind it. This document supplies the derivation, and it turns out to be short.

The five is the observer dimension. s_N is the retention of one coordinate; the observer state has five; s^5 is the joint retention and s^{-5} is the same state read backwards. The exponent was never free.

The operator is made unit-invariant by defining resolution as a quantity divided by its own uncertainty. That single move answers the two open questions that were really one question.

There is a unique resolution where the operator meets the corpus floor, $N = 5/r^3 = 40\phi^3 = 169.4427$, and it is exact.

And it is a logistic. $H_5 = 1 + \tanh(5 \ln s_N)$, which is the canonical form for a bounded quantity set by a forward rate against a reverse one. Two apparent contacts with the corpus constants are shown in section six to be golden-ratio identities carrying no information, and are demoted.

Every number below was verified symbolically. Where a result is exact it says exact, and where a promising coincidence turned out to be nothing it says that too, so no one spends a week on it.

SECTION ONE

The operator, and that its algebra holds

Written from intuition, and correct anyway. Every proposition checks.

For a positive resolution N , set the single-coordinate retention

$$s_N = N / (N + 1)$$

and compare the reciprocal fifth-power branches through their normalised imbalance. Clearing the negative branch by multiplying above and below by s^5 gives a ratio of tenth powers, and adding one renormalises it to a retention:

$$C_5(N) = (s^5 - s^{-5}) / (s^5 + s^{-5}) = (s^{10} - 1) / (s^{10} + 1)$$

$$H_5(N) = 1 + C_5(N) = 2s^{10} / (1 + s^{10}) = 2N^{10} / (N^{10} + (N+1)^{10})$$

Exact. The last form is what a symbolic algebra system returns when asked to simplify $1 + C_5$, with no intermediate steps supplied.

The four stated properties all hold. Since $N > 0$ puts s in the open unit interval, s^{10} does too, and $2t/(1+t) < 1$ exactly when $t < 1$:

Property	Statement	Status
Boundedness	$0 < H_5(N) < 1$ for $N > 0$	exact
Monotonicity	H_5 increases with N	exact, since $2t/(1+t)$ rises with t
Identity limit	$H_5 \rightarrow 1$ as $N \rightarrow \infty$	exact
Large- N expansion	$H_5 = 1 - 5/N + 5/(2N^2) + O(N^{-3})$	exact, term for term

The expansion is the one that matters, because it is where the physics is. **The unresolved fraction of a quantity is $5/N$ at leading order.** Everything else in this document is a consequence of that line.

SECTION TWO

Why the five is not a choice

The exponent was already fixed by the Passage and Record theorem, one section earlier.

The open question asked why the five observer components should act multiplicatively as a fifth power rather than through some other composition law. The answer is in the theorem that precedes it.

The Passage and Record theorem establishes that two observers may share final position and final clock reading while having accumulated different history, so an independent record coordinate is required and the minimum complete observer state is

$$O = (x, y, z, \tau, H) \quad \dim(O)_{\min} = 3 + 1 + 1 = 5$$

Now read s_N as what it is. It is the retention of **one** coordinate at resolution N: one part in $N+1$ is lost. Five coordinates retaining independently retain s^5 . So:

Object	Reading	Forced by
s_N	retention of one coordinate	definition of resolution
s^5	joint retention of the whole observer state	$\dim(O) = 5$
s^{-5}	the same state read backwards	the inverse branch
C_5	normalised imbalance, forward against inverse	passage is signed, record is not
$s^{10} = (s^5)^2$	one complete out-and-back traversal	the product of the two branches

THE EXPONENT TEN, FOR FREE

Because s^{10} is $(s^5)^2$, the tenth power is not a second assumption. It is one forward pass through the five-dimensional observer state and one return.

That is the helix, stated in retention rather than in geometry: **the phase comes back and the height does not.** The forward branch and the inverse branch are the two things the record coordinate exists to distinguish, and C_5 is precisely their imbalance.

The five is therefore derived, not posited. The operator follows from the observer-state dimension, which follows from the theorem. No step in that chain is free.

SECTION THREE

Making it invariant

Two open questions were the same question, and answering it once answers both.

The decimal-lift form carries a real problem, and the document that introduced it flagged it. The residual is not approximately proportional to the declared unit, it **is** the declared unit:

$$\epsilon = H_5(\lambda q/q_0) - 1 \approx -5 q_0 / (\lambda q)$$

Verified numerically against the exact operator to four decimal places. Which means the same operator on the same body returns whatever the unit chooses:

Declared unit q_0	Jupiter surface gravity, $\lambda = 10$	Residual ϵ
1 m/s ²	$u = 247.9$	-2.013%
1 ft/s ²	$u = 813.3$	-0.614%
1 cm/s ²	$u = 24\,790$	-0.020%
1 g (Earth)	$u = 25.28$	-19.16%
1 km/s ²	$u = 0.2479$	-99.99998%

So the Jupiter number is a statement about the metre and the second. It cannot be set beside a relativistic residual, because the two are not answering the same question.

The fix is already in the document, one subsection earlier. The fixed-resolution form carries exact scale covariance, and the proof is the fourth proposition:

$$q_p R(\alpha q, N) = \alpha \cdot q_p R(q, N)$$

That form is invariant. It is invariant because N does not depend on the magnitude of q . So the whole difficulty reduces to one question: what is N , physically, rather than numerically. And there is one answer that is unit-free by construction.

THE INVARIANT DEFINITION OF RESOLUTION

$N = q / \delta q$. The quantity divided by its own uncertainty. The number of distinguishable levels the observer actually holds.

This is invariant because a unit change scales q and δq together, so their ratio is untouched. It is physical rather than declared, because δq is a property of the measurement and not of the notation. And it makes the sampling depth literal: $\sigma = \log_{10}(q/\delta q)$ is the number of significant digits, so a tenfold lift really is one more digit.

The unresolved fraction then reads **5 $\delta q / q$, five times the relative uncertainty**. Five, because five observer coordinates. A quantity known to one part in a thousand carries a deficit of half a per cent.

This answers the first open question, on the physical definition of N , and the fourth, on the reference scale that makes the operator unit invariant. They were one question. The reference scale is the uncertainty, and the resolution is the reciprocal of the relative uncertainty.

It also disposes of λ . Once N is $q/\delta q$ there is no lift to choose: the operator is evaluated at the resolution the measurement actually has. The pair of free choices collapses to none.

SECTION FOUR

Where the operator is allowed to speak

H_5 is a correction only in the large- N regime. Below that it is a suppression.

The retention is not a small adjustment across its whole domain, and the protocol has to say where it applies:

N	$H_5(N)$	Deficit	Regime
1	0.001951	−99.8%	annihilation, not correction
5	0.278097	−72.19%	suppression
10	0.556523	−44.35%	suppression
50	0.901309	−9.87%	onset of the correction regime
169.44	0.970587	−2.94%	the floor. see section five
1 000	0.995003	−0.500%	correction
10^4	0.999500	−0.050%	correction
10^6	0.999995	−0.0005%	identity to five places

A ROW OF THE PROTOCOL THAT DOES NOT SURVIVE

The comparison table proposed applying the operator to Jupiter's compactness, $2GM/Rc^2$. That quantity is 4.03×10^{-8} , so at $\lambda = 10$ the operator returns $H_5 = 2.3 \times 10^{-64}$, which is not a residual, it is an erasure.

The idea is fine and the normalisation is not. A small dimensionless quantity has to be expressed as its own resolution before the operator sees it, and under $N = q/\delta q$ it is: compactness known to three digits gives $N = 10^3$ and a deficit of 0.5 per cent, regardless of how small the compactness itself is. **The operator should be fed resolutions, never magnitudes.**

SECTION FIVE

The join point

One resolution where Personal Relativity's deficit equals the corpus floor, in closed form.

The corpus carries an irreducible cost per completed traversal, the floor $r^3 = 0.0295085$. The operator carries an unresolved fraction $5/N$. Those are the same kind of quantity, so they can be set equal, and the resolution that does it is exact:

$$5/N = r^3 \quad \Rightarrow \quad N \dagger = 5/r^3 = 40 \varphi^3 = 169.4427$$

Exact. Since $r^3 = 1/(8\varphi^3)$, the ratio $5/r^3$ is $40\varphi^3$ with no residue. $\sigma \dagger = \log_{10} N \dagger = 2.2290$, so the join sits a little over two decades of resolution, or between two and three significant digits.

The join point is not decoration. It divides the operator's domain into two regions with different standing:

Region	Condition	What the operator may claim
Above the join	$N > 169.44$, deficit $< r^3$	Nothing. The output is quieter than the framework's own noise, so it cannot be offered as a prediction.
At the join	$N = 40\varphi^3$ exactly	The two frameworks agree on the size of the unresolved fraction. This is the only resolution at which they speak with one voice.
Below the join	$N < 169.44$, deficit $> r^3$	A claim above the floor, and therefore testable. This is where the empirical programme has to live.

WHAT THIS DOES TO THE JUPITER FIGURE

Jupiter's surface gravity at $\lambda = 10$ and $q_0 = 1 \text{ m/s}^2$ gives $N = 247.9$, which is **above the join**. Its deficit of 2.013 per cent is smaller than the framework's own floor of 2.951 per cent.

So the framework cannot claim it. Not because the arithmetic is wrong, it is exactly right, but because the framework's own resolution limit is coarser than the effect. **To make a claim the corpus licenses, the protocol needs N below 169.**

SECTION SIX**What the operator actually is, and two things it is not**

One structural identity worth having, and two apparent contacts that do not survive inspection.

The operator has a closed form that is far more revealing than the ratio of tenth powers, and it is exact:

$$H_5(N) = 1 + \tanh(5 \ln s_N) = 2 \cdot \text{logistic}(10 \ln s_N)$$

Exact, verified to 2×10^{-16} across four decades of N . The hyperbolic tangent form follows immediately once C_s is recognised as $\tanh$ of five times the log-retention of one coordinate.

This is the identity to keep, because it says what kind of object the operator is. **A hyperbolic tangent is what a bounded quantity always becomes when it is set by the competition between a forward rate and a reverse rate.** The operator is therefore a saturating response in log-retention space, five coordinates deep, and the exponent ten is the round trip. It was written down as a statement about how numbers divide over distance and it came out as the canonical form for exactly that situation.

TWO COINCIDENCES, DEMOTED

The single-coordinate retention equals r at $N = 1/\sqrt{5}$, and equals the Born coefficient $2r$ at $N = \phi$. Both are exact. **Neither is evidence.**

For any value of r whatsoever, $s_N = r$ at $N = r/(1-r)$. So the first statement carries no information about the operator; its whole content is that $r/(1-r) = 1/\sqrt{5}$, which is a property of the golden ratio standing alone. The second is the same situation, resting only on $\phi + 1 = \phi^2$. Any map of the form $x/(x+1)$ reproduces both.

They are recorded here because they look like contacts between the two frameworks and they are not. The reason is arithmetic and nothing else: an identity that holds for every value of a constant tells you nothing about that constant.

AND ONE MORE, RECORDED AS A NON-RESULT

The resolution of maximum informational leverage, where one further decade changes the retention most, sits at N^*

= **6.2825413**. That is one part in ten thousand from $2\pi = 6.2831853$, and it is not 2π .

A smooth maximum landing that close to a famous constant is the kind of thing that costs a week, and the week is not worth spending. **Coincidence, logged.**

What survives is the part that does not need the operator to have a pedigree: the exponent follows from the Passage and Record theorem, the invariant form is $N = q/\delta q$ with deficit $5\delta q/q$, and the join at $4\phi^3$ is exact. None of those three depend on where the operator came from.

RECORDED AS A NON-RESULT

The resolution of maximum informational leverage, where one further decade changes the retention most, sits at $N^* = \mathbf{6.2825413}$. That is one part in ten thousand from $2\pi = 6.2831853$, and it is not 2π .

It is written down here so that it stays written down. A smooth maximum landing that close to a famous constant is the kind of thing that costs a week, and the week is not worth spending. **Coincidence, logged.**

SECTION SEVEN

What is now open, and what closed

Four of the eight open questions are answered. The list is shorter and sharper.

Open question, as posed	Status now
The physical definition of resolution N	Closed. $N = q/\delta q$, the quantity over its own uncertainty.
Why the five components act as a fifth power	Closed. The five is $\dim(O)_m$ in from the Passage and Record theorem, and the tenth power is one out-and-back traversal.
Which quantity the operator acts upon	Closed. A resolution, never a magnitude. Section four.
What reference scale q_0 makes the operator unit invariant	Closed. $q_0 = \delta q$. Same answer as the first, same question.
How the record H should be defined per observable	Open. Total variation is the natural candidate and is what the discrete update already implements.
Whether $\kappa(\sigma)$ and ℓ_σ follow from the operator	Open, and now sharper: the operator supplies the σ dependence, so the coefficients should be read off $\partial H_5/\partial \sigma$ rather than introduced.
Whether the locked law gives useful residual structure across fields	Open, and testable only below the join. Section five.
Whether personal time can be predicted independently	Open. This remains the principal unfinished component.

The empirical programme changes shape as a result. It is no longer a matter of running the operator on bodies and reading residuals, because above the join those residuals are below the framework's own noise and below the join the normalisation has to be declared first. The programme is:

Step	What it requires
1	For each observable, state δq from the measurement, not from a unit. That fixes N with no freedom.
2	Keep only observables whose N falls below 169.44, since only those carry a claim the corpus licenses.
3	Compare the deficit $5\delta q/q$ against the observed discrepancy, with the operator locked across every body.
4	Record every failure. A residual that persists after N is raised is structure the operator does not explain, which is the outcome the operator is most useful for.

THE ONE-LINE VERSION

A quantity known to a relative precision p carries an observer-state deficit of $5p$, where the five is the dimension of the observer state and nothing else. The deficit is unit-invariant, vanishes as the measurement improves, and is a claim the framework can defend only while it stays above the floor r^3 , which fixes the working resolution at $40\phi^3$.

PROVENANCE

The operator H_5 is Joseph Shields' own. It was written down first and its exponent was accounted for afterwards, which is the ordinary order of things and is recorded only because the order is worth knowing. This document supplies that account, the invariant definition of the resolution, the domain of validity, and the join point with the corpus floor. Every algebraic claim was verified symbolically. Personal Relativity Working Formalism, Draft 0.1, 30 July 2026.